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(57) Abstract:

ABSTRACT MULTI-HAZARD PREDICTION SYSTEM USING ADAPTIVE TERRAIN GRAPH A computer-implemented method for predicting natural hazards in a terrain region comprises acquiring elevation data, partitioning the terrain region into spatial units, and constructing a terrain graph in which each spatial unit is represented as a node and graph edges represent hydrological flow relationships between neighboring spatial units. The method comprises receiving environmental input data including rainfall data and terrain-related parameters, maintaining for each spatial unit a rainfall memory state representing cumulative rainfall influence over time, and computing for each spatial unit a hazard probability based on terrain slope, the rainfall memory state, terrain condition, and hydrological flow propagation through the terrain graph. The method comprises identifying high-risk spatial units whose hazard probability exceeds a predefined threshold, adaptively refining the high-risk spatial units into smaller sub-units, and generating predictive outputs comprising a flood risk map, a landslide risk map, or a composite hazard map.[0106] where is the system state at time and is the change in sensor data. The digital twin state update enables the virtual terrain model to mirror the physical environment by incorporating incoming sensor data into the system state. [0107] Future hazard risk is estimated using the predictive function: 5 10 15 20 25 30 [0108] where is the current hazard value, is rainfall forecast, and is water flow dynamics between nodes and . The predictive function enables hazard computation engine 110 to estimate future hazard conditions based on current hazard state, anticipated rainfall, and hydrological flow patterns. [0109] Hazard computation engine 110 includes a flood simulation module that computes water accumulation and downstream flood progression using the terrain graph, rainfall input, and elevation-driven flow relationships. The flood simulation module traverses the terrain graph to propagate water accumulation values from upstream nodes to downstream nodes along directed edges. Flood related component 514 is calculated based on rainfall accumulation, upstream downstream connectivity within the terrain graph, and terrain slope. [0110] Hazard computation engine 110 includes a soil saturation

estimation module that estimates soil saturation conditions within terrain cells based on cumulative effect of rainfall and flood-related water accumulation. The soil saturation estimation module receives rainfall memory state 508 and water accumulation values from the flood simulation module to determine soil saturation levels for each spatial unit. [0111] Hazard computation engine 110 includes a landslide stress computation module that estimates landslide stress or landslide susceptibility based on the relationship among terrain slope, soil saturation state, and terrain resistance characteristics. Landslide-related component 516 is calculated based on the relationship among terrain slope, soil saturation state, and terrain resistance characteristics provided by terrain condition input 512. [0112] Hazard computation engine 110 includes a composite hazard risk estimation module that combines flood index and landslide stress values to compute composite hazard probability 518 representing joint risk of flood occurrence, landslide occurrence, or simultaneous multi-hazard interaction. In some cases, the composite hazard risk estimation module applies weighted combination of flood related component 514 and landslide-related component 516. In other cases, the composite hazard risk estimation module applies maximum value selection or probabilistic combination methods. 5 10 15 20 25 30 [0113] Alternative methods for hazard probability calculation may be employed by hazard computation engine 110. In some cases, hazard computation engine 110 applies machine learning models trained on historical hazard data to estimate hazard probabilities. In other cases, hazard computation engine 110 applies physics-based simulation models that incorporate detailed hydrological and geotechnical parameters. Hazard computation engine 110 may also apply ensemble methods that combine multiple hazard estimation approaches to generate composite hazard probability 518. In some cases, hazard computation engine 110 applies Bayesian inference methods to update hazard probabilities based on observed environmental conditions. In other cases, hazard computation engine 110 applies fuzzy logic methods to handle uncertainty in terrain and environmental parameters. [0114] FIG. 6 illustrates a predictive output visualization 600 generated by hazard visualization module 114 according to various embodiments. [0115] Output visualization 600 comprises a plurality of hazard maps generated based on computed hazard probabilities and the adaptively refined terrain representation. Hazard visualization module 114 generates a flood risk map 602 representing flood-related hazard probabilities across the terrain region. Flood risk map 602 displays spatial distribution of flood-related component values computed by hazard computation engine 110 for each spatial unit. Hazard visualization module 114 generates a landslide risk map 604 representing landslide-related hazard probabilities across the terrain region. Landslide risk map 604 displays spatial distribution of landslide-related component values computed by hazard computation engine 110 for each spatial unit. [0116] Hazard visualization module 114 generates a composite hazard map 606 representing combined flood and landslide hazard probabilities across the terrain region. Composite hazard map 606 displays spatial distribution of composite hazard probability values that combine flood-related component and landslide related component for each spatial unit. Hazard visualization module 114 generates a hazard probability map 608 representing aggregated hazard probability values across the terrain region. Hazard probability map 608 may display overall hazard probability values computed for each spatial unit based on terrain slope, rainfall memory state, terrain condition, and hydrological flow propagation through the terrain graph. [0117]Hazard visualization module 114 generates intermediate analytical outputs in addition to final composite risk maps. In some cases, hazard visualization module 114 generates flood index estimation maps that display flood index values computed by the flood simulation module for each spatial unit. In other cases, hazard visualization module 114 generates landslide stress estimation maps that display landslide stress values computed by the landslide stress computation module for each spatial unit. [0118]Output visualization 600 includes a severity legend 610 that indicates hazard severity levels associated with flood risk map 602, landslide risk map 604, composite hazard map 606, and hazard probability map 608. Severity legend 610 may use a color-coded representation to indicate different levels of hazard severity on the output maps. In some cases, severity legend 610 employs a gradient color scale ranging from low-hazard colors to high-hazard colors. In other cases, severity legend 610 employs discrete color bands corresponding to predefined hazard severity categories. The color-coded legend may assign colors such as green to low-hazard areas, yellow to moderate-hazard areas, orange to elevated-hazard areas, and red to high-hazard areas. [0119]Hazard visualization module 114 generates hazard probability maps together with histograms representing the probability distribution of hazard values across the terrain. In some cases, hazard visualization module 114 generates probability histogram outputs that display frequency distribution of hazard probability values across all spatial units of the terrain region. The probability histogram outputs may enable analysis of overall hazard distribution patterns and identification of terrain regions with concentrated hazard probability values. In other cases, hazard visualization module 114 generates cumulative distribution plots that display cumulative probability of hazard values across the terrain region. [0120]FIG. 7 illustrates a scenario simulation environment 700 of the digital twin for simulating hazard behavior under different rainfall scenarios according to various embodiments. [0121]Scenario simulation environment 700 enables predictive scenario analysis by simulating hazard behavior under varying rainfall conditions. Scenario simulation environment 700 includes a terrain region under analysis 710 representing the geographic area for which hazard simulations are

performed. Terrain region under analysis 710 corresponds to the terrain region partitioned into spatial units and represented by the terrain graph. [0122]Scenario simulation environment 700 generates a normal rainfall scenario output 702 representing predicted hazard probability values across terrain region under analysis 710 under normal rainfall conditions. Normal rainfall scenario output 702 displays hazard probability distribution when rainfall intensity corresponds to typical or average precipitation levels for the terrain region. [0123]Scenario simulation environment 700 generates a heavy rainfall scenario output 704 representing predicted hazard probability values across terrain region under analysis 710 under heavy rainfall conditions. Heavy rainfall scenario output 704 displays hazard probability distribution when rainfall intensity exceeds normal precipitation levels and corresponds to elevated rainfall events. [0124]Scenario simulation environment 700 generates an extreme rainfall scenario output 706 representing predicted hazard probability values across terrain region under analysis 710 under extreme rainfall conditions. Extreme rainfall scenario output 706 displays hazard probability distribution when rainfall intensity corresponds to severe precipitation events such as storms or monsoon conditions. [0125]Scenario simulation environment 700 includes a hazard probability scale 708 that indicates hazard probability values associated with normal rainfall scenario output 702, heavy rainfall scenario output 704, and extreme rainfall scenario output 706. Hazard probability scale 708 may display a range of probability values from low probability to high probability. In some cases, hazard probability scale 708 employs a numerical scale ranging from zero to one. In other cases, hazard probability scale 708 employs a percentage scale or a categorical scale with discrete probability levels. [0126]The digital twin environment is configured to simulate hazard behavior under different rainfall scenarios by dynamically refining the terrain graph representation based on a calculated hazard index derived from environmental parameters. As described above, hazard computation engine 110 calculates a hazard index for each terrain node based on rainfall intensity, soil moisture, and terrain slope. The digital twin environment uses the calculated hazard index to identify spatial units that require adaptive refinement under each rainfall scenario. In some cases, the digital twin environment applies different refinement thresholds for different rainfall scenarios such that more spatial units are refined under heavy rainfall and extreme rainfall conditions compared to normal rainfall conditions. [0127]The digital twin environment dynamically adjusts the terrain graph representation by subdividing high-hazard spatial units into finer-resolution sub-units as rainfall intensity increases. The dynamic refinement enables scenario simulation environment 700 to capture localized hazard behavior that may emerge under elevated rainfall conditions. In some cases, the digital twin environment maintains separate terrain graph representations for each rainfall scenario. In other cases, the digital twin environment applies incremental refinement to a base terrain graph representation as rainfall intensity parameters are adjusted. [0128]Scenario simulation environment 700 enables comparison of hazard outcomes across different rainfall scenarios. In some cases, scenario simulation environment 700 generates side-by-side visualizations of normal rainfall scenario output 702, heavy rainfall scenario output 704, and extreme rainfall scenario output 706 to enable visual comparison of hazard progression as rainfall severity increases. In other cases, scenario simulation environment 700 generates difference maps that display changes in hazard probability between rainfall scenarios. [0129]Alternative visualization formats may be employed by hazard visualization module 114. In some cases, hazard visualization module 114 generates three-dimensional terrain visualizations with hazard probability values mapped onto terrain surfaces. In other cases, hazard visualization module 114 generates time-series animations that display temporal evolution of hazard probability values as rainfall events progress. Hazard visualization module 114 may also generate contour maps that display hazard probability isolines across the terrain region. [0130]Alternative scenario types may be simulated by scenario simulation environment 700. In some cases, scenario simulation environment 700 simulates hazard behavior under varying soil moisture conditions by adjusting initial soil saturation parameters. In other cases, scenario simulation environment 700 simulates hazard behavior under different terrain condition scenarios such as varying vegetation cover or land use configurations. Scenario simulation environment 700 may also simulate compound scenarios that combine multiple environmental parameter variations to assess multi-factor hazard interactions. [0131]It is to be appreciated that the concepts, systems, and techniques sought to be protected herein are not limited to use in the example applications described herein, but rather may be useful in substantially any application where it is desired to predict, monitor, or assess natural hazards across terrain regions. The multi-hazard prediction system and associated methods disclosed herein are susceptible of industrial application in a variety of fields and industries. [0132]In some aspects, the disclosed system and method may be applied in the disaster management and emergency response industry, where government agencies, civil protection authorities, and emergency management organizations may utilize the multi-hazard prediction capabilities to support early warning systems, evacuation planning, and resource allocation during flood and landslide events. The adaptive terrain graph representation and scenario simulation capabilities may enable emergency responders to anticipate hazard progression and prepare appropriate response measures. [0133]In some aspects, the disclosed system and method may be applied in the infrastructure planning and civil engineering industry, where urban planners, transportation authorities, and construction companies may utilize the hazard probability outputs to inform site selection,

structural design, and risk mitigation strategies for roads, bridges, buildings, and other infrastructure assets located in terrain regions susceptible to flood or landslide hazards. [0134]In some aspects, the disclosed system and method may be applied in the insurance and financial services industry, where insurers and reinsurers may utilize the composite hazard maps and probability assessments to evaluate risk exposure, determine premium pricing, and support underwriting decisions for properties and assets located in hazard-prone terrain regions. [0135]In some aspects, the disclosed system and method may be applied in the environmental monitoring and climate science industry, where research institutions, meteorological agencies, and environmental consultancies may utilize the digital twin simulation environment to study hazard behavior under varying rainfall scenarios and assess the effects of climate variability on flood and landslide risk patterns. [0136]In some aspects, the disclosed system and method may be applied in the agriculture and land management industry, where farmers, forestry managers, and land conservation organizations may utilize the terrain-based hazard assessments to identify areas at risk of soil erosion, waterlogging, or slope instability, thereby supporting land use planning and soil conservation practices. [0137]While particular embodiments and applications of the present disclosure have been illustrated and described in connection with flood and landslide prediction, it is to be understood that embodiments of the disclosure are not limited to the precise construction and compositions disclosed herein and that various modifications, changes, and variations can be apparent from the foregoing descriptions without departing from the spirit and scope of the disclosure as defined in the appended claims. [0138]There are several technical advantages of the multi-hazard prediction system of the present disclosure. First, the integration of a terrain graph representation with hydrological flow relationships enables efficient propagation of hazard computations across terrain regions, allowing the system to model water flow direction and accumulation patterns based on actual terrain topography rather than relying on static analytical models. Second, the rainfall memory computation

FORM-2

THE PATENTS ACT, 1970

(39 of 1970)

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THE PATENTS RULES, 2003

COMPLETE

SPECIFICATION

(See section 10 and rule 13)

TITLE OF INVENTION

MULTI-HAZARD PREDICTION SYSTEM USING ADAPTIVE TERRAIN GRAPH

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THE FOLLOWING SPECIFICATION PARTICULARLY DESCRIBES THE INVENTION AND THE MANNER IN WHICH IT IS TO BE PERFORMED

CROSS-REFERENCE TO RELATED APPLICATIONS AND PRIORITY

[0001] The present application does not claim priority from any patent application.

PREAMBLE

[0002] The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF INVENTION

[0003] The present disclosure relates to geospatial intelligence and disaster risk prediction systems, and more particularly to a computer-implemented system and method for multi-hazard prediction using a digital twin terrain graph with adaptive spatial refinement for estimating flood and landslide probabilities across terrain regions.

BACKGROUND

[0004] The following text of background art is provided for purposes of understanding and is not an admission that any cited document or technique forms part of the prior art under applicable law.

[0005] Natural hazards such as floods and landslides pose substantial risks to human life, infrastructure, and economic stability, particularly in regions characterized by complex terrain, variable precipitation patterns, and changing environmental conditions. Accurate prediction of such hazards has become increasingly relevant as climate variability intensifies and urbanization expands into vulnerable terrain regions.

[0006] Conventional approaches to flood and landslide prediction have relied on static terrain models, threshold-based alert mechanisms, historical data analysis, or post-event monitoring. For example, US20170255692A1 discloses a digital twin simulation system for managing physical assets through continuous monitoring and real-time simulation of system behavior, but is designed primarily for industrial equipment observation and does not develop models for environmental terrain or disaster risk prediction. US20190303868A1 describes an environmental monitoring system using a distributed network of IoT sensors to monitor environmental parameters including rainfall, humidity, and temperature, but

performs data acquisition and observation without integration with a digital twin model for predictive simulation of hazards.

[0007] US20200372451A1 discloses a disaster prediction platform that compares current environmental data to historical datasets and remote sensing information to predict when a disaster event is likely to occur, but uses static analytical models without real-time digital twin simulation or dynamic terrain refinement for localized hazard prediction. WO2021102456A1 describes an environmental digital twin system that generates a virtual environmental model using GIS mapping and hydrological models to simulate flooding, but employs fixed resolution terrain representations that do not adapt computational load based on hazard risk. US20210248712A1 discloses a flood risk analysis system using rainfall, GIS mapping, and hydrological modelling data, but relies on static spatial grid models without dynamic refinement to increase precision in high-risk areas.

[0008] WO2022035678A1 describes a smart disaster monitoring platform using satellite imaging, IoT sensors, and analytical dashboards to monitor water levels and disaster-related events, but does not provide predictive simulation using a digital twin architecture. US20230123456A1 discloses an AI-based environmental risk prediction system using machine learning models to predict environmental hazards, but operates the AI prediction model separately from terrain simulation without adaptive modifications to terrain modelling and simulation resolution. US20190234561A1 describes a terrain simulation system using grid-based geographic representation, but employs uniform grid resolution across the entire terrain without capability to increase level of detail in high-risk areas. WO2019156789A1 discloses a sensor-based flood monitoring network for detecting floods and sending alerts, but provides real-time alerting without predictive modelling or digital twin simulation. US20220098765A1 describes a real-time disaster alert system using predefined thresholds to trigger alerts, but does not provide simulation of risk propagation or terrain interactions with hazards.

[0009] Existing systems generally lack adaptive terrain resolution capabilities, limited integration of digital twin technology with disaster prediction, absence of hazard-aware terrain graph models, high computational cost in large-scale

simulations due to uniform processing across entire terrain regions, limited real-time data synchronization, lack of dynamic hazard propagation modeling, and insufficient support for predictive disaster simulation. Accordingly, improved systems and methods that address one or more of these limitations are desirable.

[0010] The present invention addresses the above shortcomings of the prior art. However, the invention is entirely different from the prior art in terms of novelty and technological advancements.

OBJECT

[0011] In view of the foregoing, and without prejudice to the generality of the disclosure, it is an object of the present invention to provide. It is to be understood that the foregoing objects are illustrative and non-limiting; additional objects will be apparent from the description, examples, and claims, and need not be achieved in every embodiment.

[0012] The object of the present disclosure is to provide a computer-implemented system and method for multi-hazard prediction that overcomes the limitations of existing disaster prediction and environmental simulation systems. The system aims to enable accurate prediction of flood and landslide hazards by integrating terrain modeling, rainfall dynamics, hydrological flow propagation, and adaptive spatial refinement within a unified digital twin framework.

[0013] A further object of the present disclosure is to provide a terrain graph-based representation for modeling hydrological flow relationships between neighboring terrain units, thereby enabling simulation of water flow propagation and hazard interactions across terrain regions based on elevation gradients and environmental conditions.

[0014] Another object of the present disclosure is to provide a rainfall memory computation mechanism that tracks cumulative rainfall influence over time, thereby supporting estimation of soil moisture and saturation conditions for improved hazard probability calculation.

[0015] Yet another object of the present disclosure is to provide an integrated multi-hazard prediction framework capable of jointly estimating flood risk based

on water accumulation and downstream flood progression, and landslide risk based on terrain slope and soil saturation, within a single computational system.

[0016] A still further object of the present disclosure is to provide a probability-driven adaptive spatial refinement mechanism that dynamically adjusts the computational resolution of terrain regions based on predicted hazard probability, thereby enabling higher-resolution hazard analysis in high-risk areas while maintaining computational efficiency in lower-risk regions.

[0017] Another object of the present disclosure is to provide a digital twin-based simulation environment capable of simulating hazard behavior under different rainfall scenarios, including normal, heavy, and extreme rainfall conditions, thereby enabling predictive scenario analysis and real-time decision support for disaster preparedness and early warning applications.

[0018] For the avoidance of doubt, the foregoing objects encompass compositions and manufacturing processes suitable for adoption in jurisdictions where claims to methods of treatment of animals may be restricted; corresponding "use" or "product" claim formats are contemplated without limitation.

SUMMARY

[0019] The following summary is intended to introduce aspects of the present invention in a simplified form and does not identify all the features or the scope of the claimed invention. The summary is provided to assist in understanding the invention and is to be read in conjunction with the detailed description and accompanying drawings, claims, and abstract. One or more embodiments of the present disclosure are set forth below solely for illustrative purposes and should not be construed as limiting.

[0020] According to an aspect of the present disclosure, a computer-implemented method for predicting natural hazards in a terrain region is provided. The method comprises acquiring, by a terrain data processing module (100), elevation data corresponding to the terrain region. The method further comprises partitioning, by a grid generation module (102), the terrain region into a plurality of spatial units. The method comprises constructing, by a terrain graph construction module (104), a terrain graph (G) in which each spatial unit is represented as a node

(N) and graph edges (E) represent hydrological flow relationships between neighboring spatial units based on relative elevation differences. The method comprises receiving, by a data acquisition layer (106), environmental input data including rainfall data and terrain-related parameters associated with the spatial units. The method comprises maintaining, by a rainfall memory computation module (108), for each spatial unit, a rainfall memory state representing cumulative rainfall influence over time. The method comprises computing, by a hazard computation engine (110), for each spatial unit, a hazard probability based on terrain slope, the rainfall memory state, terrain condition, and hydrological flow propagation through the terrain graph (G). The method comprises identifying, by the hazard computation engine (110), one or more high-risk spatial units whose hazard probability exceeds a predefined threshold. The method comprises adaptively refining, by an adaptive grid refinement module (112), the identified high-risk spatial units into smaller sub-units for higher-resolution hazard analysis. The method comprises generating, by a hazard visualization module (114), predictive outputs comprising at least one of a flood risk map, a landslide risk map, or a composite hazard map based on the computed hazard probabilities and the adaptively refined terrain representation.

[0021] The foregoing method provides technical advantages including enabling integrated multi-hazard prediction through a unified terrain graph representation that captures hydrological flow relationships between spatial units. The graph-based approach facilitates efficient propagation of hazard computations across terrain regions while the rainfall memory state enables consideration of cumulative precipitation effects on soil saturation and flood potential. The adaptive refinement mechanism provides computational efficiency by concentrating processing resources on high-risk areas while maintaining lower resolution in areas of reduced hazard probability.

[0022] According to other aspects of the present disclosure, the terrain graph (G) may be represented as a directed graph in which directed edges between neighboring spatial units model downhill hydrological flow based on elevation gradients.

[0023] The directed graph representation provides technical advantages by enabling accurate modeling of water flow direction based on terrain topography. This directional modeling supports realistic simulation of downstream flood progression and upstream water accumulation patterns that influence hazard probability calculations.

[0024] According to other aspects of the present disclosure, the rainfall memory state may be updated using a temporal accumulation and decay function such that an effect of previous rainfall events is incorporated into current hazard estimation.

[0025] The temporal accumulation and decay function provides technical advantages by enabling the system to account for antecedent moisture conditions that affect soil saturation levels. This approach supports more accurate hazard estimation by recognizing that the effect of rainfall on terrain stability and flood potential diminishes over time while cumulative rainfall events may compound hazard risk.

[0026] According to other aspects of the present disclosure, the computed hazard probability may include a flood-related component based on water accumulation and downstream flood progression, and a landslide-related component based on terrain slope and soil saturation.

[0027] The dual-component hazard probability calculation provides technical advantages by enabling simultaneous assessment of multiple hazard types within a single computational framework. This integrated approach supports identification of areas where flood and landslide risks may interact or compound, providing more comprehensive hazard assessment than systems that evaluate each hazard type independently.

[0028] According to other aspects of the present disclosure, the adaptive refining may be performed for spatial units whose hazard probability exceeds the predefined threshold, while lower-risk spatial units retain an original spatial resolution.

[0029] The selective adaptive refinement provides technical advantages by optimizing computational resource allocation based on hazard probability. This

approach enables high-resolution analysis in areas where detailed hazard assessment is beneficial while avoiding unnecessary computational overhead in lower-risk regions, thereby supporting scalable hazard prediction across large terrain areas.

[0030] According to other aspects of the present disclosure, the predictive outputs may be updated iteratively in a digital twin environment as new rainfall data or terrain-condition data are received.

[0031] The iterative updating within a digital twin environment provides technical advantages by enabling continuous hazard monitoring and prediction that responds to changing environmental conditions. This approach supports real-time decision making for disaster preparedness and early warning applications by maintaining current hazard assessments as new data becomes available.

[0032] According to other aspects of the present disclosure, hydrological flow propagation may be computed using upstream and downstream node connectivity within the terrain graph (G).

[0033] The computation of hydrological flow propagation using node connectivity provides technical advantages by enabling simulation of water movement patterns across the terrain region. This graph-based flow computation supports accurate modeling of how water accumulates in downstream areas and how upstream conditions influence downstream hazard probabilities.

[0034] According to other aspects of the present disclosure, refined sub-units may inherit one or more terrain attributes from a parent spatial unit and are reevaluated at finer spatial resolution to generate localized hazard probabilities.

[0035] The inheritance of terrain attributes by refined sub-units provides technical advantages by maintaining consistency between parent and child spatial units while enabling localized hazard assessment. This hierarchical approach supports efficient refinement operations by leveraging existing terrain data while generating higher-resolution hazard predictions in areas of elevated risk.

[0036] According to other aspects of the present disclosure, a digital twin environment may be configured to simulate hazard behavior under different rainfall

scenarios by dynamically refining the terrain graph (G) representation based on a calculated hazard index derived from environmental parameters.

[0037] The scenario-based simulation capability provides technical advantages by enabling predictive analysis of hazard behavior under varying rainfall conditions including normal, heavy, and extreme rainfall scenarios. This approach supports proactive disaster preparedness planning by allowing assessment of potential hazard outcomes before actual weather events occur.

[0038] According to other aspects of the present disclosure, the flood-related component may be calculated based on rainfall accumulation, upstream-downstream connectivity within the terrain graph (G), and terrain slope, and the landslide-related component may be calculated based on a relationship among terrain slope, soil saturation state, and terrain resistance characteristics.

[0039] The detailed calculation approach for flood-related and landslide-related components provides technical advantages by incorporating multiple terrain and environmental factors into hazard probability estimation. The consideration of upstream-downstream connectivity for flood assessment and the combination of slope, saturation, and resistance characteristics for landslide assessment enables physically-informed hazard predictions that account for the distinct mechanisms underlying each hazard type.

[0040] The foregoing general description of the illustrative embodiments and the following detailed description thereof are merely exemplary aspects of the teachings of this disclosure and are not restrictive.

BRIEF DESCRIPTION OF FIGURES

[0041] The drawings appended herein illustrate exemplary embodiments of the present invention and are provided to enhance understanding of the inventive features disclosed. These figures, when read in conjunction with the detailed description, depict the operational flow, component architecture, and other mechanisms.

[0042] The illustrations are schematic in nature and serve solely to clarify the principles and functionality of the present disclosure. While particular constructions and configurations are depicted for explanatory purposes, it will be understood that

the invention is not limited to the precise implementations shown in the figures and may encompass variations without departing from the scope of protection.

[0043] FIG. 1 illustrates a block diagram of a multi-hazard prediction system for predicting natural hazards in a terrain region, according to aspects of the present disclosure.

[0044] FIG. 2 illustrates a flowchart of a method for predicting natural hazards in a terrain region, according to aspects of the present disclosure.

[0045] FIG. 3 illustrates a schematic representation of a terrain graph constructed by a terrain graph construction module, according to an embodiment.

[0046] FIG. 4 illustrates an adaptive refinement process performed by an adaptive grid refinement module, according to aspects of the present disclosure.

[0047] FIG. 5 illustrates a hazard computation pipeline implemented by a rainfall memory computation module and a hazard computation engine, according to an embodiment.

[0048] FIG. 6 illustrates a predictive output visualization generated by a hazard visualization module, according to aspects of the present disclosure.

[0049] FIG. 7 illustrates a scenario simulation environment of a digital twin for simulating hazard behavior under different rainfall scenarios, according to an embodiment.

[0050] Furthermore, in terms of the construction of the system, one or more components of the device may have been represented in the figures by conventional symbols, and the figures may show only those specific details that are pertinent to understanding the embodiments of the present invention so as not to obscure the figures with details that will be readily apparent to those of ordinary skill in the art having benefit of the description herein.

DETAILED DESCRIPTION

[0051] The following detailed description is intended to provide an in-depth understanding of the invention and its various components, configurations, and operational aspects. While specific embodiments are described with reference to the accompanying drawings, the invention is not limited to the particular forms

disclosed herein. The scope of the invention shall be interpreted broadly and in accordance with the claims appended hereto.

[0052] The terms “comprises,” “comprising,” “includes,” “including,” and similar expressions used in this description are intended to be open-ended and should be interpreted to include, but not be limited to, the mentioned elements. Unless explicitly stated otherwise, singular terms may include their plural equivalents and vice versa, depending on the context in which they appear.

[0053] Reference is now made to the accompanying figures, which illustrate example embodiments of the present invention. It should be appreciated that the figures are intended for illustration and explanatory purposes only and do not limit the scope of the invention. Similar reference numerals have been used to denote functionally similar components throughout the figures.

[0054] The embodiments described herein are presented for illustrative purposes and are subject to variations, modifications, and adaptations by those skilled in the art. Any equivalent implementations that perform substantially the same function in substantially the same manner are intended to fall within the scope of this disclosure and the claims appended hereto.

[0055] The use of the term “exemplary” in the description shall be understood to mean “serving as an example or illustration” and shall not be construed as limiting the invention to the preferred embodiments disclosed. Various other embodiments may be developed without departing from the spirit or essential characteristics of the invention. The technical principles and advantages outlined in the foregoing description should be read in conjunction with the figures and claims. It is expressly understood that while the detailed description sets out specific structural and operational aspects, the scope of protection is governed by the claims and includes all legal equivalents of the claimed subject matter. Embodiments of the disclosure are described in the following paragraphs with reference to Figures. In Figures, the same elements or elements that have the same functions are indicated by the same reference signs.

[0056] The present invention will now be described in detail with reference to the accompanying drawings illustrating exemplary embodiments.

[0057] FIG. 1 illustrates a block diagram of a multi-hazard prediction system 10 for predicting natural hazards in a terrain region according to various embodiments.

[0058] Multi-hazard prediction system 10 comprises a layered architecture including a physical environment layer, a data acquisition layer, a terrain digital twin core, a hazard computation engine, an adaptive simulation layer, and a decision support layer. Multi-hazard prediction system 10 integrates terrain modeling, rainfall dynamics, hydrological flow propagation, and adaptive spatial refinement within a unified digital twin framework to enable prediction of flood and landslide hazards.

[0059] Multi-hazard prediction system 10 includes a terrain data processing module 100 configured to acquire elevation data corresponding to the terrain region. Terrain data processing module 100 may acquire Digital Elevation Model (DEM) data representing the terrain of a study area. In some cases, terrain data processing module 100 receives elevation data from satellite sources, aerial surveys, or ground-based measurement systems. Terrain data processing module 100 may be implemented as a software module executing on a processor, a dedicated hardware circuit, a field-programmable gate array (FPGA), or a combination thereof.

[0060] Multi-hazard prediction system 10 further includes a grid generation module 102 configured to partition the terrain region into a plurality of spatial units. Grid generation module 102 partitions the terrain into a plurality of spatial grid cells, each containing elevation, slope, and related terrain attributes. In some cases, grid generation module 102 generates uniform grid cells across the terrain region, while in other cases, grid generation module 102 generates variable-sized grid cells based on terrain characteristics. Grid generation module 102 may be implemented as a software component, a hardware accelerator, or a distributed computing module.

[0061] Multi-hazard prediction system 10 includes a terrain graph construction module 104 configured to construct a terrain graph in which each spatial unit is represented as a node and graph edges represent hydrological flow relationships

between neighboring spatial units based on relative elevation differences. The terrain graph may be represented using the mathematical notation $G = (V, E)$ where V is the set of terrain nodes and E is the set of edges connecting terrain nodes. Each node in the terrain graph represents a terrain cell corresponding to a spatial unit of the terrain region.

[0062] Each node in the terrain graph contains environmental attributes represented as a node attribute tuple $N_i = (e_i, m_i, r_i, s_i)$ where e_i is elevation, m_i is soil moisture, r_i is rainfall intensity, and s_i is slope. Terrain graph construction module 104 establishes directed edges between neighboring cells based on relative elevation differences, thereby representing the natural downhill flow of water along terrain gradients. In some cases, terrain graph construction module 104 may be implemented using graph database systems, adjacency matrix representations, or linked data structures.

[0063] Multi-hazard prediction system 10 includes a data acquisition layer 106 configured to receive environmental input data including rainfall data and terrain-related parameters associated with the spatial units. Data acquisition layer 106 receives environmental input data from satellite rainfall data sources, soil moisture sensors, river level sensors, and rainfall sensors. In some cases, data acquisition layer 106 includes an environmental data collector that receives inputs from physical environment sources and a data preprocessing module that processes the collected data before passing the data to subsequent layers. Data acquisition layer 106 may receive rainfall input streams from historical datasets, real-time rainfall sensors, satellite rainfall sources, or other weather data sources. Data acquisition layer 106 may be implemented as a middleware component, a message queue system, a streaming data processor, or a combination thereof.

[0064] Multi-hazard prediction system 10 includes a rainfall memory computation module 108 configured to maintain, for each spatial unit, a rainfall memory state representing cumulative rainfall influence over time. Rainfall memory computation module 108 tracks cumulative rainfall influence over time, thereby supporting estimation of soil moisture and runoff conditions. In some cases, rainfall memory computation module 108 applies a temporal accumulation function

and a decay function to compute the rainfall memory state. Rainfall memory computation module 108 may be implemented as a time-series processing module, a state machine, or a recursive computation engine.

[0065] Multi-hazard prediction system 10 includes a hazard computation engine 110 configured to compute, for each spatial unit, a hazard probability based on terrain slope, the rainfall memory state, terrain condition, and hydrological flow propagation through the terrain graph. Hazard computation engine 110 may compute a flood-related component based on water accumulation and downstream flood progression, and a landslide-related component based on terrain slope and soil saturation. In some cases, hazard computation engine 110 combines the flood-related component and the landslide-related component to produce a composite hazard probability for each spatial unit. Hazard computation engine 110 may be implemented as a parallel processing module, a graphics processing unit (GPU) accelerated computation engine, or a distributed computing cluster.

[0066] Multi-hazard prediction system 10 includes an adaptive grid refinement module 112 configured to adaptively refine high-risk spatial units into smaller sub-units for higher-resolution hazard analysis. Adaptive grid refinement module 112 identifies terrain cells whose hazard probability exceeds a predefined threshold and automatically subdivides such high-risk cells into finer-resolution sub-cells to enable more detailed local analysis. Lower-risk cells may be maintained at coarser resolution in order to preserve computational efficiency. Adaptive grid refinement module 112 may be implemented as a hierarchical mesh refinement system, a quadtree-based subdivision module, or an octree-based spatial partitioning system.

[0067] Multi-hazard prediction system 10 includes a hazard visualization module 114 configured to generate predictive outputs comprising at least one of a flood risk map, a landslide risk map, or a composite hazard map based on the computed hazard probabilities and the adaptively refined terrain representation. Hazard visualization module 114 may generate hazard probability maps and associated probability distribution plots. The predictive outputs support disaster planning, mitigation, and early warning decision-making. Hazard visualization module 114 may be implemented as a geographic information system (GIS)

rendering module, a web-based visualization dashboard, or a desktop mapping application.

[0068] Multi-hazard prediction system 10 synchronizes real-time environmental information between actual terrain sensors and the digital twin model to enable the virtual terrain model to reflect the current environmental state. In some cases, multi-hazard prediction system 10 updates the digital twin continuously using incoming sensor data. Multi-hazard prediction system 10 may be applicable to environmental monitoring, early warning systems, climate-risk assessment, and smart infrastructure planning applications. Multi-hazard prediction system 10 may be particularly useful in regions experiencing extreme precipitation, mountainous terrain, or complex hydrological conditions where flood and landslide risk is present.

[0069] The interconnections between modules of multi-hazard prediction system 10 enable data flow from terrain data processing module 100 through grid generation module 102 and terrain graph construction module 104 to establish the terrain representation. Data acquisition layer 106 provides environmental inputs to rainfall memory computation module 108 and hazard computation engine 110. Hazard computation engine 110 receives the terrain graph from terrain graph construction module 104, the rainfall memory state from rainfall memory computation module 108, and environmental parameters from data acquisition layer 106 to compute hazard probabilities. Adaptive grid refinement module 112 receives hazard probability values from hazard computation engine 110 and provides refined terrain representations to hazard visualization module 114 for output generation.

[0070] FIG. 2 illustrates a flowchart of a method 200 for predicting natural hazards in a terrain region according to various embodiments.

[0071] Method 200 begins at a step 202 of acquiring elevation data corresponding to the terrain region. In step 202, terrain data processing module 100 acquires elevation data representing the topography of the terrain region. The elevation data may include Digital Elevation Model (DEM) data obtained from satellite sources, aerial surveys, light detection and ranging (LiDAR) systems, or ground-based measurement systems. In some cases, step 202 includes

preprocessing the elevation data to remove noise, fill data gaps, or resample the elevation data to a target resolution. In other cases, step 202 includes acquiring elevation data from multiple sources and merging the elevation data into a unified elevation dataset.

[0072] Method 200 proceeds to a step 204 of partitioning the terrain region into a plurality of spatial units. In step 204, grid generation module 102 divides the terrain region into discrete spatial units for subsequent processing. Each spatial unit may correspond to a grid cell containing elevation, slope, and terrain attributes. In some cases, step 204 generates uniform grid cells across the terrain region. In other cases, step 204 generates variable-sized grid cells based on terrain characteristics such as slope variability or elevation gradients. Step 204 may employ regular grid partitioning, triangulated irregular network (TIN) partitioning, or hexagonal grid partitioning depending on the terrain characteristics and computational requirements.

[0073] Method 200 continues to a step 206 of constructing a terrain graph in which each spatial unit is represented as a node and graph edges represent hydrological flow relationships between neighboring spatial units based on relative elevation differences. In step 206, terrain graph construction module 104 generates a terrain graph $G = (V, E)$ where each node in the set V corresponds to a spatial unit and each edge in the set E represents a hydrological flow relationship between neighboring spatial units. The terrain graph construction module 104 establishes directed edges between neighboring spatial units based on relative elevation differences, thereby modeling the natural downhill flow of water along terrain gradients. In some cases, step 206 includes computing flow direction for each spatial unit using a D8 algorithm, a D-infinity algorithm, or a multiple flow direction algorithm. In other cases, step 206 includes assigning edge weights based on slope magnitude or flow accumulation values.

[0074] Method 200 proceeds to a step 208 of receiving environmental input data including rainfall data and terrain-related parameters associated with the spatial units. In step 208, data acquisition layer 106 receives environmental input data from various sources including satellite rainfall data sources, soil moisture

sensors, river level sensors, and rainfall sensors. The environmental input data may include real-time rainfall measurements, historical rainfall records, soil type classifications, vegetation cover data, and land use information. In some cases, step 208 includes receiving streaming data from Internet of Things (IoT) sensor networks deployed across the terrain region. In other cases, step 208 includes retrieving archived environmental data from remote databases or cloud storage systems. Step 208 may include data validation, quality control, and temporal alignment of environmental input data from multiple sources.

[0075] Method 200 continues to a step 210 of maintaining, for each spatial unit, a rainfall memory state representing cumulative rainfall influence over time. In step 210, rainfall memory computation module 108 tracks cumulative rainfall influence over time, thereby supporting estimation of soil moisture and runoff conditions. The rainfall memory state captures the effect of antecedent precipitation on current terrain conditions. In some cases, step 210 applies a temporal accumulation function that aggregates rainfall values over a specified time window. In other cases, step 210 applies a decay function that reduces the influence of older rainfall events relative to more recent rainfall events. The rainfall memory state may be updated continuously as new rainfall data are received, or the rainfall memory state may be updated at discrete time intervals.

[0076] Method 200 proceeds to a step 212 of computing, for each spatial unit, a hazard probability based on terrain slope, the rainfall memory state, terrain condition, and hydrological flow propagation through the terrain graph G. In step 212, hazard computation engine 110 evaluates hazard probability for each spatial unit by combining terrain attributes with environmental conditions. Hazard computation engine 110 may compute a flood-related component based on water accumulation and downstream flood progression through the terrain graph G. Hazard computation engine 110 may compute a landslide-related component based on terrain slope and soil saturation derived from the rainfall memory state. In some cases, step 212 includes propagating hazard values through the terrain graph G using upstream and downstream node connectivity. In other cases, step 212

includes combining the flood-related component and the landslide-related component to produce a composite hazard probability for each spatial unit.

[0077] Method 200 continues to a step 214 of identifying one or more high-risk spatial units whose hazard probability exceeds a predefined threshold. In step 214, hazard computation engine 110 compares the computed hazard probability for each spatial unit against the predefined threshold. Spatial units whose hazard probability exceeds the predefined threshold are classified as high-risk spatial units. In some cases, step 214 includes applying multiple thresholds corresponding to different hazard severity levels. In other cases, step 214 includes applying separate thresholds for flood-related hazard probability and landslide-related hazard probability.

[0078] Method 200 proceeds to a step 216 of adaptively refining the identified high-risk spatial units into smaller sub-units for higher-resolution hazard analysis. In step 216, adaptive grid refinement module 112 subdivides the high-risk spatial units identified in step 214 into finer-resolution sub-units. Each sub-unit inherits terrain attributes from the parent spatial unit and is reevaluated at finer spatial resolution to generate localized hazard probabilities. Lower-risk spatial units retain an original spatial resolution to preserve computational efficiency. In some cases, step 216 employs quadtree-based subdivision in which each high-risk spatial unit is divided into four sub-units. In other cases, step 216 employs hierarchical mesh refinement with variable subdivision ratios based on hazard probability magnitude. Step 216 may be performed iteratively such that sub-units exceeding the predefined threshold are further subdivided until a minimum spatial resolution is reached.

[0079] Method 200 concludes at a step 218 of generating predictive outputs based on the computed hazard probabilities and the adaptively refined terrain representation. In step 218, hazard visualization module 114 generates predictive outputs comprising at least one of a flood risk map, a landslide risk map, or a composite hazard map based on the computed hazard probabilities and the adaptively refined terrain representation. The predictive outputs may include hazard probability maps, severity classification maps, and probability distribution plots. In some cases, step 218 includes rendering the predictive outputs as geographic

information system (GIS) layers for display on mapping applications. In other cases, step 218 includes generating alert notifications when hazard probabilities exceed predefined warning thresholds.

[0080] The predictive outputs may be updated iteratively in a digital twin environment as new rainfall data or terrain-condition data are received. In some cases, method 200 is executed continuously such that steps 208 through 218 are repeated as new environmental input data become available. In other cases, method 200 is executed at scheduled intervals or in response to detected changes in environmental conditions. The iterative updating enables multi-hazard prediction system 10 to maintain current hazard assessments that reflect evolving terrain and weather conditions.

[0081] FIG. 3 illustrates a schematic representation of a terrain graph 300 constructed by terrain graph construction module 104 according to various embodiments.

[0082] Terrain graph 300 comprises a plurality of nodes 302, each node 302 representing a spatial unit of the terrain region. Each node 302 may be associated with an elevation attribute 306 corresponding to the elevation value of the spatial unit represented by node 302. Adjacent nodes 302 are connected by directed edges 304, each directed edge 304 representing a hydrological flow relationship between neighboring spatial units based on relative elevation differences. As described above, graph edges (E) represent hydrological flow relationships between neighboring spatial units based on relative elevation differences.

[0083] Terrain graph 300 may be represented as a directed graph in which directed edges 304 between neighboring spatial units model downhill hydrological flow based on elevation gradients. Each directed edge 304 extends from an upstream node 308 to a downstream node 310 in accordance with a hydrological flow direction 312. Upstream node 308 corresponds to a spatial unit having a higher elevation value, and downstream node 310 corresponds to a spatial unit having a lower elevation value. Hydrological flow direction 312 indicates the direction of water movement from upstream node 308 toward downstream node 310 along terrain gradients.

[0084] Terrain graph construction module 104 establishes directed edges 304 between neighboring cells based on relative elevation differences to represent natural downhill flow of water along terrain gradients. The directed edge representation enables simulation of water flow propagation and hazard interactions across terrain regions based on elevation gradient and environmental conditions. Multi-hazard prediction system 10 simulates propagation of risk associated with flooding and landslides based on elevation gradient and environmental conditions through terrain graph 300.

[0085] Water flow propagation between terrain nodes may be modeled using the formula:

[0086] where F_{ij} is water flow from node i to node j, k is a flow constant, and e_i and e_j are elevation values at nodes i and j, respectively. The flow constant k may be determined based on terrain characteristics such as surface roughness, soil permeability, or vegetation cover. In some cases, the flow constant k varies across different regions of terrain graph 300 based on local terrain conditions.

[0087] Hydrological flow propagation may be computed using upstream and downstream node connectivity within terrain graph 300. Hazard computation engine 110 may traverse terrain graph 300 to propagate water accumulation values from upstream nodes 308 to downstream nodes 310 along directed edges 304. The upstream and downstream node connectivity enables hazard computation engine 110 to model how water accumulates in downstream areas and how upstream conditions influence downstream hazard probabilities.

[0088] In some cases, terrain graph 300 may be represented using alternative graph structures. Terrain graph 300 may be implemented as an adjacency matrix representation in which matrix entries indicate connectivity between nodes 302. In other cases, terrain graph 300 may be implemented as an adjacency list representation in which each node 302 maintains a list of connected neighboring nodes. Terrain graph 300 may also be implemented using graph database systems that support efficient traversal and query operations across large terrain regions.

[0089] FIG. 4 illustrates an adaptive refinement process 400 performed by adaptive grid refinement module 112 according to various embodiments.

[0090] Adaptive refinement process 400 evaluates each spatial unit of the terrain region against a hazard threshold 406 representing a predefined hazard probability value. Spatial units whose hazard probability does not exceed hazard threshold 406 are classified as low-hazard spatial units 402. Spatial units whose hazard probability exceeds hazard threshold 406 are classified as high-hazard spatial units 404.

[0091] Adaptive refining is performed for spatial units whose hazard probability exceeds the predefined threshold, while lower-risk spatial units retain an original spatial resolution. Low-hazard spatial units 402 retain an original spatial resolution to preserve computational efficiency. High-hazard spatial units 404 are adaptively subdivided into a plurality of refined sub-units 408 for higher-resolution hazard analysis.

[0092] Adaptive grid refinement module 112 may apply an adaptive resolution formula to determine the grid resolution for each spatial unit. The adaptive resolution may be expressed as:

[0093] where R_i is grid resolution at node i, H_i is the hazard probability at node i, T_h is hazard threshold 406, R_{high} is a high resolution value applied to high-hazard spatial units 404, and R_{low} is a low resolution value applied to low-hazard spatial units 402.

[0094] Each refined sub-unit 408 inherits one or more inherited terrain attributes 412 from a parent spatial unit 410. Parent spatial unit 410 corresponds to the high-hazard spatial unit 404 from which refined sub-units 408 are derived. Inherited terrain attributes 412 may include elevation values, slope values, soil type classifications, vegetation cover data, and other terrain-related parameters associated with parent spatial unit 410. Refined sub-units 408 are reevaluated at finer spatial resolution to generate localized hazard probabilities.

[0095] In some cases, adaptive grid refinement module 112 employs quadtree-based subdivision in which each high-hazard spatial unit 404 is divided into four refined sub-units 408. In other cases, adaptive grid refinement module 112 employs octree-based subdivision for three-dimensional terrain representations. Adaptive

grid refinement module 112 may also employ triangular mesh refinement in which high-hazard spatial units 404 are subdivided into smaller triangular elements.

[0096] In some cases, adaptive refinement process 400 is performed iteratively such that refined sub-units 408 whose hazard probability exceeds hazard threshold 406 are further subdivided until a minimum spatial resolution is reached. In other cases, adaptive refinement process 400 applies multiple hazard thresholds corresponding to different refinement levels, enabling graduated resolution increases based on hazard probability magnitude.

[0097] Alternative refinement strategies may be employed by adaptive grid refinement module 112. In some cases, adaptive grid refinement module 112 applies gradient-based refinement in which spatial units exhibiting high spatial gradients in hazard probability are refined regardless of absolute hazard probability values. In other cases, adaptive grid refinement module 112 applies feature-based refinement in which spatial units containing terrain features such as stream channels, ridgelines, or slope breaks are refined to capture localized hazard behavior. Adaptive grid refinement module 112 may also apply temporal refinement in which spatial resolution is increased during periods of elevated rainfall or changing environmental conditions.

[0098] FIG. 5 illustrates a hazard computation pipeline 500 implemented by rainfall memory computation module 108 and hazard computation engine 110 according to various embodiments.

[0099] Hazard computation pipeline 500 receives environmental inputs and terrain data to compute hazard probabilities for each spatial unit of the terrain region. Hazard computation pipeline 500 includes a rainfall input stream 502 that provides rainfall data to rainfall memory computation module 108. Rainfall input stream 502 may receive rainfall data from historical datasets, real-time rainfall sensors, satellite rainfall sources, or other weather data sources.

[0100] Rainfall memory computation module 108 applies a temporal accumulation function 504 and a decay function 506 to compute a rainfall memory state 508 for each spatial unit. Temporal accumulation function 504 aggregates rainfall values over a specified time window to capture cumulative rainfall

influence. Decay function 506 reduces the influence of older rainfall events relative to more recent rainfall events. The rainfall memory state 508 represents cumulative rainfall influence over time and is updated using temporal accumulation function 504 and decay function 506 such that an effect of previous rainfall events is incorporated into current hazard estimation. In some cases, temporal accumulation function 504 applies weighted summation of rainfall values over a sliding time window. In other cases, decay function 506 applies exponential decay to reduce the contribution of older rainfall measurements.

[0101] Hazard computation engine 110 receives rainfall memory state 508 from rainfall memory computation module 108, a terrain slope input 510, and a terrain condition input 512. Terrain slope input 510 provides slope values derived from elevation data for each spatial unit. Terrain condition input 512 provides terrain-related parameters including soil type classifications, vegetation cover data, and terrain resistance characteristics associated with each spatial unit.

[0102] Hazard computation engine 110 computes a flood-related component 514 and a landslide-related component 516 for each spatial unit. Flood-related component 514 is based on water accumulation and downstream flood progression. Landslide-related component 516 is based on terrain slope and soil saturation. Hazard computation engine 110 combines flood-related component 514 and landslide-related component 516 to produce a composite hazard probability 518 for each spatial unit.

[0103] Hazard computation engine 110 calculates a hazard risk index for each terrain node using the formula:

[0104] where H_i is the hazard index for node i, r_i is rainfall intensity, m_i is soil moisture, s_i is terrain slope, and α, β, and γ are weighting coefficients. The weighting coefficients α, β, and γ determine the contribution of rainfall intensity, soil moisture, and terrain slope, respectively, to the hazard index calculation. In some cases, the weighting coefficients are determined based on historical hazard data or terrain characteristics of the study area. In other cases, the weighting coefficients are adjusted dynamically based on current environmental conditions.

[0105] The digital twin state update is performed using the formula:

[0106] where S_t is the system state at time t and ΔD is the change in sensor data. The digital twin state update enables the virtual terrain model to mirror the physical environment by incorporating incoming sensor data into the system state.

[0107] Future hazard risk is estimated using the predictive function:

[0108] where H_t is the current hazard value, R_t is rainfall forecast, and F_{ij} is water flow dynamics between nodes i and j. The predictive function enables hazard computation engine 110 to estimate future hazard conditions based on current hazard state, anticipated rainfall, and hydrological flow patterns.

[0109] Hazard computation engine 110 includes a flood simulation module that computes water accumulation and downstream flood progression using the terrain graph, rainfall input, and elevation-driven flow relationships. The flood simulation module traverses the terrain graph to propagate water accumulation values from upstream nodes to downstream nodes along directed edges. Flood-related component 514 is calculated based on rainfall accumulation, upstream-downstream connectivity within the terrain graph, and terrain slope.

[0110] Hazard computation engine 110 includes a soil saturation estimation module that estimates soil saturation conditions within terrain cells based on cumulative effect of rainfall and flood-related water accumulation. The soil saturation estimation module receives rainfall memory state 508 and water accumulation values from the flood simulation module to determine soil saturation levels for each spatial unit.

[0111] Hazard computation engine 110 includes a landslide stress computation module that estimates landslide stress or landslide susceptibility based on the relationship among terrain slope, soil saturation state, and terrain resistance characteristics. Landslide-related component 516 is calculated based on the relationship among terrain slope, soil saturation state, and terrain resistance characteristics provided by terrain condition input 512.

[0112] Hazard computation engine 110 includes a composite hazard risk estimation module that combines flood index and landslide stress values to compute composite hazard probability 518 representing joint risk of flood occurrence, landslide occurrence, or simultaneous multi-hazard interaction. In some cases, the

composite hazard risk estimation module applies weighted combination of flood-related component 514 and landslide-related component 516. In other cases, the composite hazard risk estimation module applies maximum value selection or probabilistic combination methods.

[0113] Alternative methods for hazard probability calculation may be employed by hazard computation engine 110. In some cases, hazard computation engine 110 applies machine learning models trained on historical hazard data to estimate hazard probabilities. In other cases, hazard computation engine 110 applies physics-based simulation models that incorporate detailed hydrological and geotechnical parameters. Hazard computation engine 110 may also apply ensemble methods that combine multiple hazard estimation approaches to generate composite hazard probability 518. In some cases, hazard computation engine 110 applies Bayesian inference methods to update hazard probabilities based on observed environmental conditions. In other cases, hazard computation engine 110 applies fuzzy logic methods to handle uncertainty in terrain and environmental parameters.

[0114] FIG. 6 illustrates a predictive output visualization 600 generated by hazard visualization module 114 according to various embodiments.

[0115] Output visualization 600 comprises a plurality of hazard maps generated based on computed hazard probabilities and the adaptively refined terrain representation. Hazard visualization module 114 generates a flood risk map 602 representing flood-related hazard probabilities across the terrain region. Flood risk map 602 displays spatial distribution of flood-related component values computed by hazard computation engine 110 for each spatial unit. Hazard visualization module 114 generates a landslide risk map 604 representing landslide-related hazard probabilities across the terrain region. Landslide risk map 604 displays spatial distribution of landslide-related component values computed by hazard computation engine 110 for each spatial unit.

[0116] Hazard visualization module 114 generates a composite hazard map 606 representing combined flood and landslide hazard probabilities across the terrain region. Composite hazard map 606 displays spatial distribution of composite hazard probability values that combine flood-related component and landslide-

related component for each spatial unit. Hazard visualization module 114 generates a hazard probability map 608 representing aggregated hazard probability values across the terrain region. Hazard probability map 608 may display overall hazard probability values computed for each spatial unit based on terrain slope, rainfall memory state, terrain condition, and hydrological flow propagation through the terrain graph.

[0117] Hazard visualization module 114 generates intermediate analytical outputs in addition to final composite risk maps. In some cases, hazard visualization module 114 generates flood index estimation maps that display flood index values computed by the flood simulation module for each spatial unit. In other cases, hazard visualization module 114 generates landslide stress estimation maps that display landslide stress values computed by the landslide stress computation module for each spatial unit.

[0118] Output visualization 600 includes a severity legend 610 that indicates hazard severity levels associated with flood risk map 602, landslide risk map 604, composite hazard map 606, and hazard probability map 608. Severity legend 610 may use a color-coded representation to indicate different levels of hazard severity on the output maps. In some cases, severity legend 610 employs a gradient color scale ranging from low-hazard colors to high-hazard colors. In other cases, severity legend 610 employs discrete color bands corresponding to predefined hazard severity categories. The color-coded legend may assign colors such as green to low-hazard areas, yellow to moderate-hazard areas, orange to elevated-hazard areas, and red to high-hazard areas.

[0119] Hazard visualization module 114 generates hazard probability maps together with histograms representing the probability distribution of hazard values across the terrain. In some cases, hazard visualization module 114 generates probability histogram outputs that display frequency distribution of hazard probability values across all spatial units of the terrain region. The probability histogram outputs may enable analysis of overall hazard distribution patterns and identification of terrain regions with concentrated hazard probability values. In

other cases, hazard visualization module 114 generates cumulative distribution plots that display cumulative probability of hazard values across the terrain region.

[0120] FIG. 7 illustrates a scenario simulation environment 700 of the digital twin for simulating hazard behavior under different rainfall scenarios according to various embodiments.

[0121] Scenario simulation environment 700 enables predictive scenario analysis by simulating hazard behavior under varying rainfall conditions. Scenario simulation environment 700 includes a terrain region under analysis 710 representing the geographic area for which hazard simulations are performed. Terrain region under analysis 710 corresponds to the terrain region partitioned into spatial units and represented by the terrain graph.

[0122] Scenario simulation environment 700 generates a normal rainfall scenario output 702 representing predicted hazard probability values across terrain region under analysis 710 under normal rainfall conditions. Normal rainfall scenario output 702 displays hazard probability distribution when rainfall intensity corresponds to typical or average precipitation levels for the terrain region.

[0123] Scenario simulation environment 700 generates a heavy rainfall scenario output 704 representing predicted hazard probability values across terrain region under analysis 710 under heavy rainfall conditions. Heavy rainfall scenario output 704 displays hazard probability distribution when rainfall intensity exceeds normal precipitation levels and corresponds to elevated rainfall events.

[0124] Scenario simulation environment 700 generates an extreme rainfall scenario output 706 representing predicted hazard probability values across terrain region under analysis 710 under extreme rainfall conditions. Extreme rainfall scenario output 706 displays hazard probability distribution when rainfall intensity corresponds to severe precipitation events such as storms or monsoon conditions.

[0125] Scenario simulation environment 700 includes a hazard probability scale 708 that indicates hazard probability values associated with normal rainfall scenario output 702, heavy rainfall scenario output 704, and extreme rainfall scenario output 706. Hazard probability scale 708 may display a range of probability values from low probability to high probability. In some cases, hazard

probability scale 708 employs a numerical scale ranging from zero to one. In other cases, hazard probability scale 708 employs a percentage scale or a categorical scale with discrete probability levels.

[0126] The digital twin environment is configured to simulate hazard behavior under different rainfall scenarios by dynamically refining the terrain graph representation based on a calculated hazard index derived from environmental parameters. As described above, hazard computation engine 110 calculates a hazard index for each terrain node based on rainfall intensity, soil moisture, and terrain slope. The digital twin environment uses the calculated hazard index to identify spatial units that require adaptive refinement under each rainfall scenario. In some cases, the digital twin environment applies different refinement thresholds for different rainfall scenarios such that more spatial units are refined under heavy rainfall and extreme rainfall conditions compared to normal rainfall conditions.

[0127] The digital twin environment dynamically adjusts the terrain graph representation by subdividing high-hazard spatial units into finer-resolution sub-units as rainfall intensity increases. The dynamic refinement enables scenario simulation environment 700 to capture localized hazard behavior that may emerge under elevated rainfall conditions. In some cases, the digital twin environment maintains separate terrain graph representations for each rainfall scenario. In other cases, the digital twin environment applies incremental refinement to a base terrain graph representation as rainfall intensity parameters are adjusted.

[0128] Scenario simulation environment 700 enables comparison of hazard outcomes across different rainfall scenarios. In some cases, scenario simulation environment 700 generates side-by-side visualizations of normal rainfall scenario output 702, heavy rainfall scenario output 704, and extreme rainfall scenario output 706 to enable visual comparison of hazard progression as rainfall severity increases. In other cases, scenario simulation environment 700 generates difference maps that display changes in hazard probability between rainfall scenarios.

[0129] Alternative visualization formats may be employed by hazard visualization module 114. In some cases, hazard visualization module 114 generates three-dimensional terrain visualizations with hazard probability values

mapped onto terrain surfaces. In other cases, hazard visualization module 114 generates time-series animations that display temporal evolution of hazard probability values as rainfall events progress. Hazard visualization module 114 may also generate contour maps that display hazard probability isolines across the terrain region.

[0130] Alternative scenario types may be simulated by scenario simulation environment 700. In some cases, scenario simulation environment 700 simulates hazard behavior under varying soil moisture conditions by adjusting initial soil saturation parameters. In other cases, scenario simulation environment 700 simulates hazard behavior under different terrain condition scenarios such as varying vegetation cover or land use configurations. Scenario simulation environment 700 may also simulate compound scenarios that combine multiple environmental parameter variations to assess multi-factor hazard interactions.

[0131] It is to be appreciated that the concepts, systems, and techniques sought to be protected herein are not limited to use in the example applications described herein, but rather may be useful in substantially any application where it is desired to predict, monitor, or assess natural hazards across terrain regions. The multi-hazard prediction system and associated methods disclosed herein are susceptible of industrial application in a variety of fields and industries.

[0132] In some aspects, the disclosed system and method may be applied in the disaster management and emergency response industry, where government agencies, civil protection authorities, and emergency management organizations may utilize the multi-hazard prediction capabilities to support early warning systems, evacuation planning, and resource allocation during flood and landslide events. The adaptive terrain graph representation and scenario simulation capabilities may enable emergency responders to anticipate hazard progression and prepare appropriate response measures.

[0133] In some aspects, the disclosed system and method may be applied in the infrastructure planning and civil engineering industry, where urban planners, transportation authorities, and construction companies may utilize the hazard probability outputs to inform site selection, structural design, and risk mitigation

strategies for roads, bridges, buildings, and other infrastructure assets located in terrain regions susceptible to flood or landslide hazards.

[0134] In some aspects, the disclosed system and method may be applied in the insurance and financial services industry, where insurers and reinsurers may utilize the composite hazard maps and probability assessments to evaluate risk exposure, determine premium pricing, and support underwriting decisions for properties and assets located in hazard-prone terrain regions.

[0135] In some aspects, the disclosed system and method may be applied in the environmental monitoring and climate science industry, where research institutions, meteorological agencies, and environmental consultancies may utilize the digital twin simulation environment to study hazard behavior under varying rainfall scenarios and assess the effects of climate variability on flood and landslide risk patterns.

[0136] In some aspects, the disclosed system and method may be applied in the agriculture and land management industry, where farmers, forestry managers, and land conservation organizations may utilize the terrain-based hazard assessments to identify areas at risk of soil erosion, waterlogging, or slope instability, thereby supporting land use planning and soil conservation practices.

[0137] While particular embodiments and applications of the present disclosure have been illustrated and described in connection with flood and landslide prediction, it is to be understood that embodiments of the disclosure are not limited to the precise construction and compositions disclosed herein and that various modifications, changes, and variations can be apparent from the foregoing descriptions without departing from the spirit and scope of the disclosure as defined in the appended claims.

[0138] There are several technical advantages of the multi-hazard prediction system of the present disclosure. First, the integration of a terrain graph representation with hydrological flow relationships enables efficient propagation of hazard computations across terrain regions, allowing the system to model water flow direction and accumulation patterns based on actual terrain topography rather than relying on static analytical models. Second, the rainfall memory computation

mechanism that tracks cumulative rainfall influence over time provides improved hazard estimation by accounting for antecedent moisture conditions that affect soil saturation levels, recognizing that the effect of rainfall on terrain stability and flood potential diminishes over time while cumulative rainfall events may compound hazard risk. Third, the adaptive grid refinement mechanism provides computational efficiency by concentrating processing resources on high-risk areas while maintaining lower resolution in areas of reduced hazard probability, thereby enabling scalable hazard prediction across large terrain areas without uniform computational overhead. Fourth, the dual-component hazard probability calculation that combines flood-related and landslide-related components within a single computational framework enables simultaneous assessment of multiple hazard types and supports identification of areas where flood and landslide risks may interact or compound. Fifth, the digital twin environment with scenario simulation capabilities enables predictive analysis of hazard behavior under varying rainfall conditions including normal, heavy, and extreme rainfall scenarios, supporting proactive disaster preparedness planning by allowing assessment of potential hazard outcomes before actual weather events occur. Sixth, the iterative updating capability within the digital twin environment enables continuous hazard monitoring and prediction that responds to changing environmental conditions, supporting real-time decision making for disaster preparedness and early warning applications. These technical advantages collectively contribute to a comprehensive solution for multi-hazard prediction that is both computationally efficient and adaptable to varying terrain and environmental conditions.

[0139] It will be appreciated that the above-detailed description, along with FIGs, is provided to illustrate the architecture and operation of exemplary embodiments of the present invention. Various modifications and enhancements can be made without departing from the scope of the invention. All such variations, as would be recognized by those skilled in the art, are intended to be within the scope of the present disclosure, which is defined by the claims that follow.

[0140] It will be appreciated by persons skilled in the art that while the invention has been described with reference to specific embodiments and

accompanying drawings, numerous modifications, substitutions, variations, and equivalents are possible without departing from the spirit or scope of the present invention. The use of singular terms such as “a,” “an,” and “the” is not intended to limit the disclosed elements to a single instance unless explicitly stated, and such terms should be interpreted to include plural forms as applicable. Likewise, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having,” and their grammatical variants are intended to be open-ended and non-limiting, and should be interpreted as meaning “including but not limited to.” Any enumerated listing of components, features, or elements does not imply that the items are mutually exclusive unless expressly stated.

[0141] It is further understood that where features, characteristics, or elements are described in connection with a Markush group or a disjunctive phrase (e.g., “A or B”), the invention includes all possible combinations and sub-combinations thereof unless explicitly excluded. Thus, a statement referring to “at least one of A, B, and C” should be interpreted to mean any one or more of A, B, and C, individually or in any combination. The language used in this specification is selected primarily for clarity and illustrative purposes and should not be construed to limit the inventive scope unless specifically recited in the claims. Accordingly, the foregoing description of embodiments should be regarded as illustrative and not restrictive, with the scope of the invention being defined solely by the appended claims and their legal equivalents

[0142] It will be recognized that various features, elements, and combinations thereof described herein may be desirably adapted for alternative implementations or other applications. Many such modifications, substitutions, enhancements, or equivalents may become apparent to those skilled in the art upon reading this disclosure and are considered to fall within the scope and spirit of the present invention. The invention is not limited to the precise configurations or exemplary embodiments set forth in this specification, and various alternatives may be used without departing from the intended objectives. The claims, and not the detailed description, shall define the legal scope of protection afforded by the present disclosure.

[0143] In interpreting the specification and claims, all terms should be construed in the broadest reasonable manner consistent with the context and the understanding of those skilled in the art. The use of terms such as "comprises," "comprising," "includes," "including," or "has" should be interpreted to be non-exclusive and not limited to the stated elements alone. Moreover, references to "one embodiment," "an embodiment," or "in certain embodiments" are not meant to imply that such features are required or exclusive, and any feature described in connection with one embodiment may be used in combination with other features or embodiments unless clearly prohibited or contextually inconsistent.

[0144] It should be further appreciated that reference throughout the specification to features, operations, or components using singular terms should not be construed as excluding a plurality thereof unless the context expressly indicates otherwise. Similarly, features described in the context of grouped or listed elements (e.g., A, B, and C) should be understood to encompass individual, multiple, or all combinations of said elements. Where ranges are given, all intermediate values and subranges are understood to be disclosed as if specifically recited. The scope of the invention should therefore be construed as inclusive of all such combinations and logical extensions as would be appreciated by a person of ordinary skill in the relevant technical field

[0145] In another embodiment, the above disclosure is a description of the invention and is not intended to limit the scope of the invention. Other variations and modifications of the above-described embodiment shall be apparent to those skilled in the art and are intended to fall within the scope of the invention as defined in the following claims.

Dated This ..... day of May 2026

Kuldeep

(Kuldeep Singh)
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CLAIMS

WE CLAIM:

1. A computer-implemented method for predicting natural hazards in a terrain region, the method comprising:

acquiring, by a terrain data processing module (100), elevation data corresponding to the terrain region;

partitioning, by a grid generation module (102), the terrain region into a plurality of spatial units;

constructing, by a terrain graph construction module (104), a terrain graph (G) in which each spatial unit is represented as a node (N) and graph edges (E) represent hydrological flow relationships between neighboring spatial units based on relative elevation differences;

receiving, by a data acquisition layer (106), environmental input data including rainfall data and terrain-related parameters associated with the spatial units;

maintaining, by a rainfall memory computation module (108), for each spatial unit, a rainfall memory state representing cumulative rainfall influence over time;

computing, by a hazard computation engine (110), for each spatial unit, a hazard probability based on terrain slope, the rainfall memory state, terrain condition, and hydrological flow propagation through the terrain graph (G);

identifying, by the hazard computation engine (110), one or more high-risk spatial units whose hazard probability exceeds a predefined threshold;

adaptively refining, by an adaptive grid refinement module (112), the identified high-risk spatial units into smaller sub-units for higher-resolution hazard analysis; and

generating, by a hazard visualization module (114), predictive outputs comprising at least one of a flood risk map, a landslide risk map, or a composite hazard map based on the computed hazard probabilities and the adaptively refined terrain representation.

2. The method of claim 1, wherein the terrain graph (G) is represented as a directed graph in which directed edges between neighboring spatial units model downhill hydrological flow based on elevation gradients.

3. The method of claim 1, wherein the rainfall memory state is updated using a temporal accumulation and decay function such that an effect of previous rainfall events is incorporated into current hazard estimation.

4. The method of claim 1, wherein the computed hazard probability includes:

a flood-related component based on water accumulation and downstream flood progression; and

a landslide-related component based on terrain slope and soil saturation.

5. The method of claim 1, wherein the adaptive refining is performed for spatial units whose hazard probability exceeds the predefined threshold, while lower-risk spatial units retain an original spatial resolution.

6. The method of claim 1, wherein the predictive outputs are updated iteratively in a digital twin environment as new rainfall data or terrain-condition data are received.

7. The method of claim 1, wherein hydrological flow propagation is computed using upstream and downstream node connectivity within the terrain graph (G).

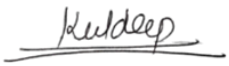
8. The method of claim 1, wherein refined sub-units inherit one or more terrain attributes from a parent spatial unit and are reevaluated at finer spatial resolution to generate localized hazard probabilities.

9. The method of claim 1, wherein a digital twin environment is configured to simulate hazard behavior under different rainfall scenarios by dynamically refining the terrain graph (G) representation based on a calculated hazard index derived from environmental parameters.

10. The method of claim 4, wherein the flood-related component is calculated based on rainfall accumulation, upstream-downstream connectivity within the terrain graph (G), and terrain slope, and wherein the landslide-related

component is calculated based on a relationship among terrain slope, soil saturation state, and terrain resistance characteristics.

Dated This ..... day of May 2026



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ABSTRACT

MULTI-HAZARD PREDICTION SYSTEM USING ADAPTIVE TERRAIN GRAPH

A computer-implemented method for predicting natural hazards in a terrain region comprises acquiring elevation data, partitioning the terrain region into spatial units, and constructing a terrain graph in which each spatial unit is represented as a node and graph edges represent hydrological flow relationships between neighboring spatial units. The method comprises receiving environmental input data including rainfall data and terrain-related parameters, maintaining for each spatial unit a rainfall memory state representing cumulative rainfall influence over time, and computing for each spatial unit a hazard probability based on terrain slope, the rainfall memory state, terrain condition, and hydrological flow propagation through the terrain graph. The method comprises identifying high-risk spatial units whose hazard probability exceeds a predefined threshold, adaptively refining the high-risk spatial units into smaller sub-units, and generating predictive outputs comprising a flood risk map, a landslide risk map, or a composite hazard map.

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